Lab experiment 2

Aim

To implement a basic finite state automaton (FSA) in Python that recognizes the pattern of strings ending with 'ab' over the alphabet {a, b}.

Algorithm

1. Start the program.
2. Define the states of the automaton: q0 (start state), q1 (last symbol read was 'a') and q2 (last two symbols read were 'ab').
3. Define the transition table that maps each (state, input symbol) pair to the next state.
4. Set q0 as the start state and q2 as the accepting (final) state.
5. Read the input string one character at a time and move to the next state using the transition table.
6. After the last character, check whether the current state is the accepting state.
7. If it is, display Accepted; otherwise display Rejected.
8. Test the automaton with several sample strings and display the results.
9. Stop the program.

Code :

```
# Finite State Automaton to accept strings ending with 'ab' over {a, b}
# States: q0 (start), q1 (last char was 'a'), q2 (last two chars were 'ab' - accepting)
transitions = {
    ("q0", "a"): "q1", ("q0", "b"): "q0",
    ("q1", "a"): "q1", ("q1", "b"): "q2",
    ("q2", "a"): "q1", ("q2", "b"): "q0",
}
start_state = "q0"
accepting_states = {"q2"}

def fsa_accepts(string):
    state = start_state
    for ch in string:
        if (state, ch) not in transitions:
            return False
```

```
        state = transitions[(state, ch)]
    return state in accepting_states

test_strings = ["ab", "aab", "bab", "abab", "a", "ba", "abb", "bbb", ""]
for s in test_strings:
    result = "Accepted" if fsa_accepts(s) else "Rejected"
    print(f"'{s}' -> {result}")
```

Output\

NLP 2.py - C:\Users\M. JAHANARA JASMIN\OneDrive\ドキュメント\NLP 2.py (3.13.12)

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```
def accepts(s):
    state = 0

    for ch in s:
        if state == 0:
            state = 1 if ch == 'a' else 0
        elif state == 1:
            state = 2 if ch == 'b' else (1 if ch == 'a' else 0)
        elif state == 2:
            state = 1 if ch == 'a' else 0

    return state == 2

s = input("Enter a string containing a and b: ")

if accepts(s):
    print("String accepted")
else:
    print("String rejected")
```

```
>>>
========= RESTART: C:\Users\M. JAHANARA JASMIN\One
=====
Enter a string containing a and b: abbab
String accepted
>>>
```

Result

Thus, the Python program to implement a finite state automaton that recognizes strings ending with 'ab' was executed successfully, and the strings were correctly accepted or rejected.